

Analyzing Hallucination Patterns in Retrieval-Augmented Financial Question Answering: A Preliminary Study

Bhalchandra Shinde, Piyush Daga, Anish Kelkar, Rituraj Singh

Retrieval-augmented generation (RAG) grounds large language models (LLMs) in external documents, yet models still produce claims that contradict or exceed the retrieved evidence—a failure mode with acute consequences in finance, where a single wrong figure misleads. We study *when and how* RAG systems hallucinate on financial QA using the open-source subset of **FinanceBench** (150 annotated questions over 84 SEC filings). This preliminary report contributes: (i) an expanded literature review situating our work across RAG methods, hallucination evaluation, and financial NLP; (ii) an exploratory analysis showing that FinanceBench answers are overwhelmingly *abstractive*—only 2% appear verbatim in the cited evidence—over long, table-heavy 10-K filings, precisely the regime where grounded generation is hard; and (iii) a simple baseline pipeline (BM25 + GPT-3.5-turbo) evaluated with RAGAS faithfulness and token-level F1/Exact Match. Our results establish a working end-to-end system and quantify the retrieval bottleneck that drives both abstention and the “grounded-but-wrong” errors we will taxonomize in the final paper.

1 Introduction

Retrieval-augmented generation (Lewis et al. 2020) has become the dominant recipe for grounding LLM outputs in verifiable external knowledge. Yet a growing body of work shows that even with relevant context in the prompt, models generate statements that are unsupported by—or contradict—the retrieved passages (Ji et al. 2023; Huang et al. 2023). In *financial* question answering this is especially costly: answers are exact numeric values drawn from dense regulatory filings, and a single fabricated or misattributed figure produces a confidently wrong answer.

We investigate three research questions:

RQ1. How frequently do different RAG pipeline configurations hallucinate on financial QA?

RQ2. Which hallucination *types* dominate in finance—numerical, entity, reasoning, or unsupported extrapolation?

RQ3. Do larger / instruction-tuned LLMs hallucinate less given the same retrieved context?

This report establishes the foundation for answering them: a literature review (Section 2), an exploratory analysis of FinanceBench that motivates the difficulty (Section 3), and a working baseline pipeline with preliminary faithfulness and accuracy metrics (Section 4). We close with the roadblocks this early evidence surfaces (Section 5).

Contributions.

Bhalchandra Shinde: repository/infrastructure, environment, data pipeline, exploratory data analysis, and literature-review notes for Lewis et al. (2020) and Islam et al. (2023). Anish Kelkar: document chunking, baseline RAG pipeline, evaluation harness (RAGAS + F1/EM), and results tabulation. Piyush Daga and Rituraj Singh: remaining related-work papers and paper writing. All members contributed to the final draft.

2 Related Work

We organize related work into three themes: RAG methods and retrieval, hallucination detection and evaluation, and financial NLP/QA.

RAG methods and retrieval. RAG was introduced by Lewis et al. (2020), coupling a DPR retriever with a BART generator and framing knowledge as *parametric* (model weights) versus *non-parametric* (an editable document index) memory; it is the architectural baseline our pipelines instantiate. Two of its findings bear directly on our setting. First, RAG answers correctly even when the gold answer appears in *no* retrieved passage verbatim (11.8% of Natural Questions cases), where a purely extractive reader scores zero—the same *abstractive* regime we observe in FinanceBench, where only 2% of gold answers are verbatim in the evidence (Section 3). Second, swapping the document index updates the model’s world knowledge with no retraining, underscoring that retrieval quality—not generator memory—governs what a RAG system can ground, which is precisely the bottleneck our results expose. The two retrieval families we compare are defined by BM25 (Robertson and Zaragoza 2009), a probabilistic term-matching function that is our baseline retriever, and DPR (Karpukhin et al. 2020), which learns dense question/passage encoders and beats BM25 on most open-domain QA—motivating our planned dense pipeline. That dense pipeline embeds chunks with Sentence-BERT (Reimers and Gurevych 2019), a siamese-BERT model that produces semantically comparable sentence vectors, the encoder behind our MiniLM index. REALM (Guu et al. 2020) pre-trains the retriever jointly with a masked LM, and Fusion-in-Decoder (Izacard and Grave 2021) instead fuses many retrieved passages in the decoder; both integrate

retrieval more deeply than our prompt-based setup and mark upgrade paths we may explore. Self-RAG (Asai et al. 2024) trains a model to decide *when* to retrieve and to critique its own generations with reflection tokens—directly relevant to cutting the unsupported answers we aim to reduce. Query rewriting (Ma et al. 2023) expands or reformulates the query before retrieval and is exactly the mechanism of our planned “enhanced” pipeline. The recent survey of Gao et al. (2023) organizes these advances and motivates our controlled comparison of retrieval strategies.

Hallucination detection and evaluation. Ji et al. (2023) and Huang et al. (2023) survey hallucination in NLG and LLMs respectively and provide the vocabulary (intrinsic vs. extrinsic, faithfulness vs. factuality) our taxonomy refines for finance. For measurement, RAGAS (Es et al. 2024) supplies reference-free metrics—faithfulness, answer relevancy, context precision—that we adopt as our primary hallucination signal. SelfCheckGPT (Manakul, Liusie, and Gales 2023) detects hallucination without external resources by sampling several responses and measuring their mutual consistency, while FActScore (Min et al. 2023) decomposes a generation into atomic facts and scores each against a source—two complementary detection lenses we consider for the final error analysis. RAGTruth (Niu et al. 2024) contributes a word-level hallucination corpus and taxonomy that we will use as supervision for the (optional) automatic classifier, and the RGB benchmark (J. Chen et al. 2024) evaluates LLM robustness to noisy retrieval—the “wrong context” condition our full-corpus retriever deliberately induces. We additionally report token-level F1/EM following the SQuAD protocol (Rajpurkar et al. 2016) to capture correctness that faithfulness alone misses, and note RAGBench (Friel, Belyi, and Sanyal 2024) as a cross-domain resource for future comparison.

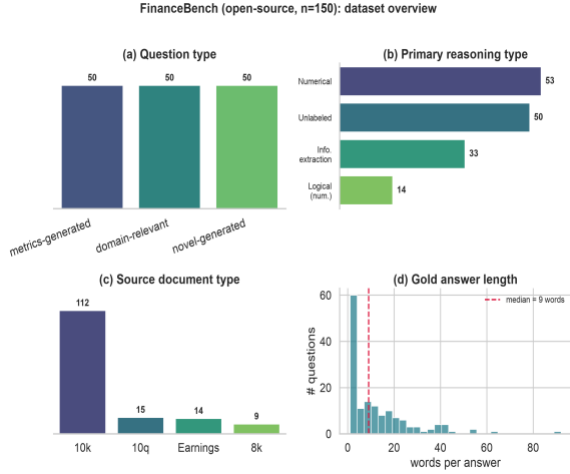
Financial NLP and QA. FinanceBench (Islam et al. 2023) is our primary benchmark: 10,231 open-book questions over 361 real 10-K/10-Q/8-K/earnings filings from 40 US companies, of which a human-reviewed 150-question subset (the open-source release) is our evaluation set. Its own experiments motivate our study in three ways. (i) 66% of its questions require *numerical reasoning* rather than extraction—our EDA independently recovers this numeric skew (Section 3). (ii) Even the best realistic configuration (GPT-4-Turbo, long context) reaches only ~79% accuracy against an 85% oracle ceiling, and confident *hallucination* is a named failure theme—the phenomenon we set out to quantify and taxonomize. (iii) They report that stronger models

tend to *refuse* when uncertain while weaker ones answer wrong; our GPT-3.5 baseline likewise abstains rather than fabricate (Section 4), and they find placing context *before* the question is critical—a design choice our prompt template adopts. Prior financial QA datasets establish that numerical reasoning over hybrid table/text is the core difficulty our EDA also finds: FinQA (Z. Chen et al. 2021) pairs expert-written questions with executable reasoning programs over earnings-report tables; ConvFinQA (Z. Chen et al. 2022) extends this to multi-turn conversational chains; and TAT-QA (Zhu et al. 2021) targets questions that jointly span a table and its surrounding text—exactly the table+prose evidence our chunker must preserve. On the model side, BloombergGPT (Wu et al. 2023), a 50B model pre-trained on a large financial corpus, and FinGPT (Yang, Liu, and Wang 2023), an open finance-LLM framework, show the appetite for finance-specialized models but evaluate generation quality rather than isolating *RAG* hallucination, which is our focus. Our open-source generator comparison will use Mistral-7B-Instruct (Jiang et al. 2023), a strong 7B open model, as the RQ3 counterpart to GPT-3.5.

3 Data and Exploratory Analysis

Dataset. We use the open-source subset of FinanceBench (Islam et al. 2023): 150 QA pairs spanning 32 companies and 84 source filings (fiscal years 2015–2024) across 9 GICS sectors. Question types are balanced by design (50 metrics-generated, 50 domain-relevant, 50 novel-generated). Each record carries a gold answer and an evidence span, letting us judge groundedness directly.

Findings. Figure 1 summarizes the analysis (full notebook and 22 figures in the repository). Four properties make this a stress test for grounded generation: (1) *Long, tabular context*—source filings have a median of 128 pages and evidence spans are 69% table-heavy, so retrieval operates over large, noisy documents. (2) *Short, numeric answers*—the median gold answer is 9 words and 84% are numeric, so a single wrong figure flips a correct answer to wrong. (3) *Abstractive answers*—only 2% of gold answers appear *verbatim* in the evidence (mean extractability 0.27); the model must *synthesize* a value, not copy it, which is the core hallucination risk. (4) *Modest lexical overlap*—median question–evidence cosine similarity is 0.47, so retrievers must bridge a vocabulary gap, and 46% of questions require numerical computation or aggregation. These observations justify measuring faithfulness (RAGAS) alongside F1/EM and motivate our four-category hallucination taxonomy.



FinanceBench (open-source, $n=150$) overview: (a) balanced question types, (b) numeric reasoning dominates, (c) 10-K filings dominate the corpus, (d) gold answers are short.

4 Preliminary Method and Results

Baseline pipeline. For each question we build a BM25 index (Robertson and Zaragoza 2009) over the *entire* chunk corpus (all 84 filings, fixed 512-token chunks), retrieve the top-3 chunks, and prompt GPT-3.5-turbo (temperature 0) to answer using only that context, instructing it to say “I don’t know” when the answer is absent. We index the full corpus on purpose: a retriever that can surface the *wrong* filing is exactly the setting where hallucination appears, which is what we study. Outputs follow a fixed schema so the same evaluation harness scores every future pipeline.

Evaluation. We report RAGAS (Es et al. 2024) faithfulness, answer relevancy, and context precision (LLM judge: GPT-3.5-turbo; local sentence-transformer embeddings), plus SQuAD-style token F1 and Exact Match (Rajpurkar et al. 2016) against gold answers. Because gold answers are short and formatted many ways (\$1,577 vs. 1577.00), EM is expected to be low; we therefore also report F1 on the answered subset.

Preliminary baseline results on FinanceBench (all 150 questions). Generator and RAGAS judge: GPT-3.5-turbo at temperature 0; RAGAS embeddings are local sentence-transformers.

Pipeline	Model	Faithfulness	Answer Rel.	Context Prec.	EM	F1
baseline_bm25	gpt-3.5-turbo	0.102	0.211	0.534	0.000	0.083

Interpretation. The baseline pairs a plain BM25 retriever with GPT-3.5-turbo. BM25 surfaces the correct filing for only 43% of questions, so the generator—instructed to answer only from context—abstains on roughly two-thirds, answering 47 of 150.

This caution is deliberate and safer than fabrication, but it caps coverage: token F1 is 0.083 over all questions and 0.261 on the answered subset, with exact match at zero because short numeric answers are written many ways (\$1,577 vs. 1577.00) and rarely match after normalization. RAGAS reflects the same bottleneck: faithfulness (0.102) and answer relevancy (0.211) are low—an abstention has nothing to ground and does not address the question—while context precision (0.534) shows retrieval places a useful chunk near the top only about half the time. Read together, the picture is a retrieval-bound baseline: the generator rarely fabricates, but weak sparse retrieval leaves most questions unanswered. Closing that gap with dense and query-rewrite retrieval—and surfacing the answered-but-unsupported cases—is the focus of the next phase.

5 Roadblocks and Next Steps

The preliminary run isolates the key bottleneck: sparse full-corpus retrieval often fails to surface the correct filing, so the model either abstains or answers from wrong context. This directly shapes our plan: (1) dense retrieval (FAISS + sentence embeddings) and query rewriting to raise retrieval recall (RQ1); (2) an open-source generator (Mistral-7B (Jiang et al. 2023)) under the same retrieval config to test RQ3; and (3) manual annotation of 50 failure cases into our four-type taxonomy (numerical, entity, reasoning, unsupported extrapolation) to answer RQ2. The main risks are OpenAI API cost control (run once at temperature 0) and the fidelity of PDF table extraction, which flattens tables into linear text and is a known limitation we will quantify.

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